



# TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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## FIRST REVISION EXAMINATION, JANUARY - 2025

### BUSINESS MATHEMATICS AND STATISTICS

Time Allowed : 3.00 Hours]

[Max. Marks : 90

#### PART - I

20 x 1 = 20

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. If  $\frac{a_1}{x} + \frac{b_1}{y} = c_1$ ,  $\frac{a_2}{x} + \frac{b_2}{y} = c_2$ ,  $\Delta_1 = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ ;  $\Delta_2 = \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}$ ;  $\Delta_3 = \begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}$  then (x, y) is

(a)  $\left(\frac{\Delta_2}{\Delta_1}, \frac{\Delta_3}{\Delta_1}\right)$

(b)  $\left(\frac{\Delta_3}{\Delta_1}, \frac{\Delta_2}{\Delta_1}\right)$

(c)  $\left(\frac{\Delta_1}{\Delta_2}, \frac{\Delta_1}{\Delta_3}\right)$

(d)  $\left(\frac{-\Delta_1}{\Delta_2}, \frac{-\Delta_1}{\Delta_3}\right)$

2.  $\Gamma\left(\frac{3}{2}\right)$

(a)  $\sqrt{\pi}$

(b)  $\frac{\sqrt{\pi}}{2}$

(c)  $2\sqrt{\pi}$

(d)  $\frac{3}{2}$

3.  $\int \frac{2x+3}{\sqrt{x^2+3x+2}} dx$  is

(a)  $\sqrt{x^2+3x+2} + c$

(b)  $2\sqrt{x^2+3x+2} + c$

(c)  $\log(x^2+3x+2) + c$

(d)  $\frac{2}{3}(x^2+3x+2)^{\frac{3}{2}} + c$

4. Area bounded by  $y = |x|$  between the limits 0 and 2 is

(a) 1sq.units

(b) 3 sq.units

(c) 2 sq.units

(d) 4 sq.units

5. The demand and supply functions are given by  $D(x) = 16 - x^2$  and  $S(x) = 2x^2 + 4$  are under perfect competition, then the equilibrium price x is

(a) 2

(b) 3

(c) 4

(d) 5

6. The order and degree of the differential equation  $\sqrt{\frac{d^2y}{dx^2}} = \sqrt{\frac{dy}{dx} + 5}$  are respectively

(a) 2 and 3

(b) 3 and 2

(c) 2 and 1

(d) 2 and 2

7. The general solution of the differential equation  $\frac{dy}{dx} = \cos x$  is

(a)  $y = \sin x + 1$

(b)  $y = \sin x - 2$

(c)  $y = \cos x + c$ , c is an arbitrary constant

(d)  $y = \sin x + c$ , c is an arbitrary constant

8. For the given data find the value of  $\Delta^3 y_0$  is

|   |    |    |    |    |
|---|----|----|----|----|
| x | 5  | 6  | 9  | 11 |
| y | 12 | 13 | 15 | 18 |

(a) 1

(b) 0

(c) 2

(d) -1

9. If c is a constant then  $\Delta c =$

(a) c

(b)  $\Delta$

(c)  $\Delta^2$

(d) 0

10. For the given points  $(x_0, y_0)$  and  $(x_1, y_1)$  the Lagrange's formula is

(a)  $y(x) = \frac{x-x_1}{x_0-x_1}y_0 + \frac{x-x_0}{x_1-x_0}y_1$  (b)  $y(x) = \frac{x_1-x}{x_0-x_1}y_0 + \frac{x-x_0}{x_1-x_0}y_1$  (c)  $y(x) = \frac{x-x_1}{x_0-x_1}y_1 + \frac{x-x_0}{x_1-x_0}y_0$  (d)  $y(x) = \frac{x_1-x}{x_0-x_1}y_1 + \frac{x-x_0}{x_1-x_0}y_0$

11. The rank of the diagonal matrix

$$\begin{pmatrix} 1 & & & & \\ & 2 & & & \\ & -3 & & & \\ & & 0 & & \\ & & & 0 & \\ & & & & 0 \end{pmatrix}$$

- (a) 0 (b) 2 (c) 3 (d) 5

12. For the system of equations  $x + 2y + 3z = 1$ ,  $2x + y + 3z = 2$ ,  $5x + 5y + 9z = 4$

- (a) there is only one solution (b) there exists infinitely many solutions  
(c) there is no solution (d) None of these

13. If  $f(x)$  is a continuous function and  $a < c < b$ , then  $\int_a^c f(x) dx + \int_c^b f(x) dx$  is

- (a)  $\int_a^b f(x) dx - \int_a^c f(x) dx$  (b)  $\int_a^c f(x) dx - \int_a^b f(x) dx$  (c)  $\int_a^b f(x) dx$  (d) 0

14.  $\int_0^1 \sqrt{x^4(1-x)^2} dx$  is

- (a) 1/12 (b) -7/12 (c) 7/12 (d) -1/12

15. The profit of a function  $p(x)$  is maximum when

- (a)  $MC - MR = 0$  (b)  $MC = 0$  (c)  $MR = 0$  (d)  $MC + MR = 0$

16. The demand and supply function of a commodity are  $D(x) = 25 - 2x$  and  $S(x) = \frac{10+x}{4}$  then the equilibrium price  $P_0$  is

- (a) 5 (b) 2 (c) 3 (d) 10

17. The solution of the differential equation  $\frac{dy}{dx} = \frac{y}{x} + \frac{f(\frac{y}{x})}{f(\frac{y}{x})}$  is

- (a)  $f(\frac{y}{x}) = kx$  (b)  $x f(\frac{y}{x}) = k$  (c)  $f(\frac{y}{x}) = ky$  (d)  $y f(\frac{y}{x}) = k$

18. The particular integral of the differential equation is  $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 2e^{4x}$

- (a)  $\frac{x^2 e^{4x}}{2!}$  (b)  $\frac{e^{4x}}{2!}$  (c)  $x^2 e^{4x}$  (d)  $x e^{4x}$

19. If  $\begin{vmatrix} 3k & 6 \\ 3 & 1 \end{vmatrix} = 0$  then value of  $k$  is

- (a) 0 (b) 2 (c) 3 (d) 18

20.  $\nabla^n y_n =$

- (a)  $y_n - y_{n-1}$  (b)  $y_n - y_{n-1} + y_{n+2}$  (c)  $y_n - 2y_{n-1} + y_{n+2}$  (d) none

PART - II

1. Answer any 7 questions

7 x 2 = 14

2. Each question carries 2 marks

3. Question number 30 is compulsory

21. Find the rank of the matrix  $A = \begin{pmatrix} 1 & -3 & 4 & 7 \\ 9 & 1 & 2 & 0 \end{pmatrix}$

22. Parithi is either sad (S) or happy (H) each day. If he is happy in one day, he is sad on the next day by four times out of five. If he is sad on one day, he is happy on the next day by two times out of three. Over a long run, what are the chances that Parithi is happy on any given day?

23. Integrate with respect to  $x$ :  $2 \cos x - 3 \sin x + 4 \sec^2 x - 5 \operatorname{cosec}^2 x$

24. Evaluate:  $\Gamma\left(\frac{9}{2}\right)$

25. Find the area bounded by  $y = 4x + 3$  with x-axis between the lines  $x = 1$  and  $x = 4$ .

26. If  $MR = 14 - 6x + 9x^2$ , find the demand function.

27. Find the differential equation of the following:  $xy = c^2$

28. Solve:  $(1-x)dy - (1+y)dx = 0$

29. Find  $\Delta \log x$

30. What is maximin criteria

## PART - III

1. Answer any 7 questions

7 x 3 = 21

2. Each question carries 3 marks

3. Question number 40 is compulsory

31. A total of ₹8,600 was invested in two accounts. One account earned  $4\frac{3}{4}\%$  annual interest and the other earned  $6\frac{1}{2}\%$  annual interest. If the total interest for one year was ₹431.25, how much was invested in each account? (Use determinant method).

32. Show that the equations  $x - 4y + 7z = 14$ ,  $3x + 8y - 2z = 13$ ,  $7x - 8y + 26z = 5$  are inconsistent.

33. Integrate with respect to  $x$ :  $x^5 e^{x^2}$

34. Integrate with respect to  $x$ :  $e^{3x} \left[ \frac{3x-1}{9x^2} \right]$

35. When the Elasticity function is  $\frac{E_y}{E_x} = \frac{x}{x-2}$ . Find the function when  $x = 6$  and  $y = 16$ .

36. Find the differential equation of the family of parabola with foci at the origin and axis along the x-axis.

37. Solve:  $\frac{dy}{dx} + \frac{3x^2}{1+x^3}y = \frac{1+x^2}{1+x^3}$

38. From the following table find the missing value

| X    | 2    | 3    | 4    | 5 | 6    |
|------|------|------|------|---|------|
| f(x) | 45.0 | 49.2 | 54.1 | - | 67.4 |

39. Prove that  $f(4) = f(3) + \Delta f(2) + \Delta^2 f(1) + \Delta^3 f(1)$  taking '1' as the interval of differencing.

40. Solve:  $\frac{d^2y}{dx^2} + 9y = 0$

## PART - IV

7 x 5 = 35

1. Answer all the questions

2. Each question carries 5 marks

41. a) Investigate for what values of 'a' and 'b' the following system of equations

$x + y + z = 6$ ,  $x + 2y + 3z = 10$ ,  $x + 2y + az = b$  have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

(OR)

b) A new transit system has just gone into operation in Chennai. Of those who use the transit system this year, 30% will switch over to using metro train next year and 70% will continue to use the transit system. Of those who use metro train this year, 70% will continue to use metro train next year and 30% will switch over to the transit system. Suppose the population of Chennai city remains constant and that 60% of the commuters use the transit system and 40% of the commuters use metro train this year.

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(i) What percent of commuters will be using the transit system after one year?

(ii) What percent of commuters will be using the transit system in the long run?

42. a) A total of ₹8,500 was invested in three interest earning accounts. The interest rates were 2%, 3% and 6% if the total simple interest for one year was ₹380 and the amount invested at 6% was equal to the sum of the amounts in the other two accounts, then how much was invested in each account? (use Cramer's rule).

(OR)

b) Evaluate  $\int_1^2 \frac{1}{(x+1)(x+2)} dx$

43. a) Evaluate:  $\int_0^3 \frac{x dx}{\sqrt{x+1} + \sqrt{5x+1}}$

(OR)

b) Evaluate the integral as the limit of the sum:  $\int_1^3 (2x+3) dx$

44. a) The demand and supply function of a commodity are  $p_d = 18 - 2x - x^2$  and  $p_s = 2x - 3$ . Find the consumer's surplus and producer's surplus at equilibrium price.

(OR)

b) Elasticity of a function  $\frac{E_y}{E_x}$  is given by  $\frac{E_y}{E_x} = \frac{-7x}{(1-2x)(2+3x)}$ . Find the function when  $x = 2, y = \frac{3}{8}$ .

45. a) Sketch the graph  $y = |x+3|$  and evaluate  $\int_{-6}^0 |x+3| dx$

(OR)

- b) The marginal cost of production of a firm is given by  $C'(x) = 5 + 0.13x$ , the marginal revenue is given by  $R'(x) = 18$  and the fixed cost is ₹ 120. Find the profit function.

46. a) Solve the homogeneous differential equations:  $(y^2 - 2xy)dx = (x^2 - 2xy)dy$

(OR)

- b) Using interpolation estimate the output of a factory in 1986 from the following data

| Year                 | 1974 | 1978 | 1982 | 1990 |
|----------------------|------|------|------|------|
| Output in 1000 tones | 25   | 60   | 80   | 170  |

47. a) Using appropriate interpolation formula find the number of students whose weight is between 60 and 70 from the data given below

| Weight in lbs   | 0-40 | 40-60 | 60-80 | 80-100 | 100-120 |
|-----------------|------|-------|-------|--------|---------|
| No. of students | 250  | 120   | 100   | 70     | 50      |

(OR)

- b) From the following table obtain a polynomial of degree y in x.

| X | 1 | 2  | 3 | 4  | 5 |
|---|---|----|---|----|---|
| Y | 1 | -1 | 1 | -1 | 1 |